

A trace-norm vertex that is not a partial permutation

Ziang Ni (ziangni-sys)

10 October 2026

Conjecture 00000007151. The source asserts that the extremal body of the trace norm in Schatten p has vertices at partial permutation matrices. With the usual norm-unit-body and unsigned partial-permutation conventions, this is false already for real 1×1 matrices. Neither language imposes entrywise positivity or a dimension restriction.

Actual matrix norm. For a real matrix A , its singular values are the nonnegative square roots of the eigenvalues of $A^T A$; its trace norm is their sum. Take the one-dimensional real matrix space $M_1(\mathbb{R})$. For $A = [a]$, actual matrix multiplication gives $A^T A = [a^2]$. Its sole eigenvalue is a^2 : multiplication on a vector $[v]$ is $[a^2 v]$, and a nonzero eigenvector forces the eigenvalue to be a^2 . Thus its sole singular value is $\sqrt{a^2} = |a|$, and

$$\|[a]\|_1 = |a|, \quad B = \{A \in M_1(\mathbb{R}) : \|A\|_1 \leq 1\} = \{[a] : -1 \leq a \leq 1\}.$$

This is the actual trace-norm unit body. It is a line segment, so the word “vertex” is unambiguous here. Choosing $p = 1$ already suffices; in fact every one-dimensional Schatten norm has the same unit body.

Theorem 1. *The matrix $V = [-1]$ is an exposed vertex of B , but is not a standard partial permutation matrix.*

Proof. Consider the nonzero matrix linear functional $\ell([a]) = -a$. For every $[a] \in B$, one has $\ell([a]) \leq 1$, with equality if and only if $a = -1$. Therefore its maximizing face on B is exactly $\{V\}$, so V is exposed. For completeness, suppose $V = tA + (1-t)C$ with $A, C \in B$ and $0 < t < 1$. Writing $A = [a]$ and $C = [c]$, we get

$$0 = t(a + 1) + (1 - t)(c + 1).$$

Both summands are nonnegative and their multipliers are positive; hence $a = c = -1$. This also proves extremality directly.

A standard partial permutation matrix has entries in $\{0, 1\}$, with at most one 1 in each row and each column. In dimension one the only such matrices are $[0]$ and $[1]$. The entry -1 of V is neither 0 nor 1, proving the claimed exclusion. $\square$

Scope and formal verification. The example uses the usual unsigned partial-permutation definition, not a modified definition allowing negative signs. The statement contains no positivity condition that would remove V . The Lean project uses actual `Matrix Unit Unit` $\mathbb{R}$, actual transpose/multiplication and vector action, the uniquely computed Gram eigenvalue, and its singular-value trace norm. It proves the singleton maximizing face of an actual linear functional, membership in `mathlib`’s `extremePoints`, and failure of the full zero-one row/column partial-permutation conditions. The final existential theorem gives the counterexample. Lean 4.19.0 and `mathlib` commit `c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned; the printed axiom audits contain only the permitted standard axioms.